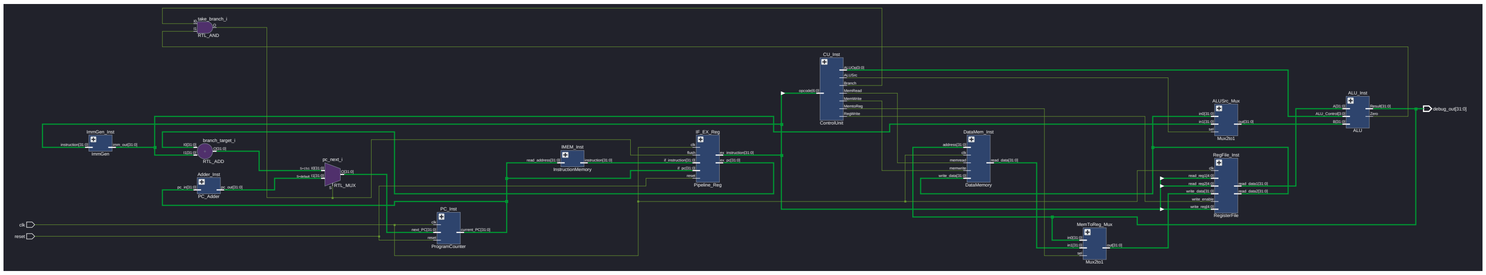
take_branch_i
I0
I1
O
RTL_AND
ImmGen_Inst
instruction[31:0]
imm_out[31:0]
ImmGen
branch_target_i
I0[31:0]
I1[31:0]
O[31:0]
RTL_ADD
Adder_Inst
pc_in[31:0]
pc_out[31:0]
PC_Adder
pc_next_i
S=1'b1 I0[31:0]
S=default I1[31:0]
O[31:0]
S
RTL_MUX
clk
reset
PC_Inst
clk
next_PC[31:0]
reset
current_PC[31:0]
ProgramCounter
IMEM_Inst
read_address[31:0]
instruction[31:0]
InstructionMemory
IF_EX_Reg
clk
flush
if_instruction[31:0]
if_pc[31:0]
reset
ex_instruction[31:0]
ex_pc[31:0]
Pipeline_Reg
CU_Inst
opcode[6:0]
ALUOp[3:0]
ALUSrc
Branch
MemRead
MemWrite
MemtoReg
RegWrite
ControlUnit
DataMem_Inst
address[31:0]
clk
memread
memwrite
write_data[31:0]
read_data[31:0]
DataMemory
ALUSrc_Mux
in0[31:0]
in1[31:0]
sel
out[31:0]
Mux2to1
RegFile_Inst
clk
read_reg1[4:0]
read_reg2[4:0]
write_data[31:0]
write_enable
write_reg[4:0]
read_data1[31:0]
read_data2[31:0]
RegisterFile
MemToReg_Mux
in0[31:0]
in1[31:0]
sel
out[31:0]
Mux2to1
ALU_Inst
A[31:0]
ALU_Control[3:0]
B[31:0]
Result[31:0]
Zero
ALU
debug_out[31:0]